FABIAN ALVAREZ-PRIMO, PhD

MATERIALS ENGINEER & OPERATIONS MANAGER | HIGH-VOLUME MATERIAL FLOW | SAFETY, QUALITY & CONTINUOUS IMPROVEMENT

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Summary

Metallurgical and materials engineer (B.S., Ph.D., UTEP) with deep expertise in failure analysis, materials characterization, and manufacturing hardware. Has deliberately built a second core strength: leading people under pressure. Took on Amazon operations to lead large teams where quality, safety and deadlines don't give. Now takes extreme ownership of a **60+ associate** high-volume material-flow operation: **99.7% quality compliance**, a **0.0% serious incident rate** for 9+ months, DPMO **29% better than goal**, and an **82% throughput increase**, driven by custom engineered Python/SQL/JS tooling built to find the bottleneck. Brings both to Starship: understands the hardware and the specs behind the material, and leads the team that keeps it flowing to the build.

Technical Skills

- **Operations & Material Flow:** receipt-to-delivery flow design, inventory and location accuracy, labor and capacity planning, lean and standard work, root cause and corrective action, KPI design and dashboards
- **Quality & Safety Systems:** CLIA/COLA, cGMP, ISO 9001, ASTM; controlled documentation, chain of custody, audit readiness; OSHA enforcement and safety-barrier programs
- **Metallurgy & Materials:** physical metallurgy and alloys, metals processing, hands-on metal casting, corrosion, mechanical behavior of materials, shelf-life and storage-sensitive materials
- **Failure Analysis & Testing:** SEM fractography, EDS, XRD, XRF, Raman/FTIR; tensile, compression, and creep testing to ASTM; TGA, DMA; test-method development
- **Manufacturing & Design:** additive manufacturing and 3D printer hardware, injection molding, extrusion; SolidWorks, Fusion 360, AutoCAD; SOPs and work instructions
- **Software & Data:** Python (pandas, scikit-learn, Streamlit), SQL, R, MATLAB, JavaScript (browser automation), Google Apps Script, Excel / Google Sheets (advanced), Tableau
- **Languages:** English, Spanish (native)

Professional Experience

Operations Manager II, L5 | Amazon | Savannah, GA | 2024 – Present

Chose high-volume operations to lead large teams in a high-stakes, high-pressure environment: hard hourly departure deadlines, heavy equipment, and zero tolerance on safety.

- Own Outbound Dock operations (Container Management Outbound Flow); prior scope covered Amazon Robotics Floor, transshipment, vendor returns, and shipping sort. Lead a **60+ associate** team across a compressed 4-day schedule at **99.7% quality compliance**.
- Raised dock rate **82% (220 → 400+ units/labor hrs)** and cut recirculation from **5% to below 2%**. Treated the dock as an engineering system: instrumented it, found the chute-level constraints, and redesigned staffing and material paths around them.
- Built a container-to-physical-location staging audit tool that maps material to its location, reducing missed packages and DPMO risk.
- Engineered a three-layer flow-visibility system (browser-automation scraper of live WIP data, a real-time analytics layer for root causes, staffing ratios, DPMO and hourly throughput, and a staffing/rate tool), replacing manual workflows and enabling proactive labor deployment.
- Sustained a **0.0% Serious Incident Rate for 9+ consecutive months**; run daily barrier sweeps and safety-reporting mechanisms on shift.
- Beat the DPMO goal by **29% (5,291 vs. 7,500)** and held AR COS penetration at **92.6–94.4% vs. a 90% goal**.
- Stood up Vendor Returns operations from scratch: wrote the process, set the standard work, and trained the team.

- Developed **15 Process Assistants** (up to **9 concurrently**) and **8 new leaders** into full role ownership.

Account Executive | DISH Network | 2023 – 2024

Chose a quota-driven, customer-facing role to build communication, persuasion, and composure under pressure.

- Won a company-wide award for the highest contracts sold during training; promoted to Account Executive.
- Built Tableau analytics to drive customer-retention strategy.

Post-doctoral Research Fellow | Texas Tech University Health Sciences Center | El Paso, TX | 2022 – 2022

- Ran postmortem human brain biobanking operations and clinical research in neuropsychiatry with the Southwest Brain Bank and El Paso County Coroner's Office, covering specimen intake, processing, controlled storage, and chain-of-custody documentation.
- Led a stem-cell reprogramming project under **cGMP-compliant** protocols and verified the integrity of stored tissue through follow-up culture testing.
- Owned statistical analysis (R, Python) across concurrent studies and coordinated deliverables across clinical, molecular, and neuroscience teams; co-authored *Frontiers in Psychiatry* (2023).

Clinical Laboratory Scientist & Supervisor | West Texas Pain Institute | El Paso, TX | 2021 – 2022

- Supervised a team of **9** that scaled sample processing **150%**, averaging **1,300+ patient samples per month**, under full **CLIA and COLA** compliance and audit readiness.
- Operated and maintained LC/MS, immunoassay, and DNA sequencing instrumentation; enforced OSHA protocols and ran the LIS (Stratus DX) and EMR data systems.

Doctoral Research Assistant | Inspired Materials & Stem Cell Tissue Engineering Lab, UTEP | El Paso, TX | 2017 – 2020

- Ran lab operations across **5+ concurrent projects**: equipment scheduling, reagent procurement, SOPs, and data management.
- Improved 3D printer manufacturing precision **30%** through hands-on hardware modifications; trained **30+ students** on SEM and 3D printing.
- Co-authored a failure analysis study of additively manufactured specimens (*Journal of Failure Analysis and Prevention*, 2019); characterized a conductive composite to ASTM and **ISO 9001** standards.

R&D Engineer Intern | Climax Molybdenum | Sahuarita, AZ | 2015 – 2015

- Developed ASTM test methods for a refractory ceramic in leaching-vessel service, led an initiative to extend its service life, and wrote the material-handling SOP.

Undergraduate Research Assistant | W.M. Keck Center for 3D Innovation, UTEP | 2014 – 2015

- Tested tensile, compression, and creep properties of aerospace-grade ABS/SEBS composites with the U.S. Air Force and America Makes; characterized fracture surfaces with SEM and XRD.

Education

Ph.D., Materials Science & Engineering | The University of Texas at El Paso | 2020 | GPA 3.91

B.S., Metallurgical & Materials Engineering | The University of Texas at El Paso | 2015

Publications & Honors

7 peer-reviewed publications, including failure analysis and materials characterization. Tau Beta Pi · Alpha Sigma Mu · Alpha Chi · Society for Biomaterials (President, 2 terms)